

Computational Search for a Perfect Cuboid: Algorithms, Near-Miss Analysis, and Scaling Feasibility

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Abstract

The *perfect cuboid problem* — whether a rectangular box exists whose three edges, three face diagonals, and space diagonal are all integers — has remained open for over three centuries. Despite exhaustive searches extending to edge bounds exceeding 10^{13} , no solution has been found, yet no proof of non-existence is known. We present a systematic computational investigation implementing four distinct search algorithms: exhaustive brute force with S_3 -symmetry reduction, a modular arithmetic sieve exploiting quadratic residue tables for 25 primes, a parametric family generator covering the Saunderson, Euler, and Bremner families, and a novel Pythagorean triple intersection method that reformulates Euler brick discovery as triangle finding in the Pythagorean graph. Our search identifies 69 Euler bricks with edges up to 5,000 and 517 parametric-family bricks with edges up to $\sim 10^{10}$, none of which are perfect cuboids. The novel algorithm achieves empirical $O(n^{1.30})$ scaling — a $780\times$ speedup over brute force — while the modular sieve eliminates $>99.999\%$ of candidates before exact checking. A chi-squared test on the near-miss gap distribution ($\chi^2 = 4.45$, $\text{dof} = 9$) shows consistency with uniformity, providing no statistical evidence for a nearby solution. We discuss implications for non-existence and identify the Brauer–Manin obstruction on the associated K3 surface as the most promising path to a theoretical resolution.

1 Introduction

A *perfect cuboid* is a rectangular parallelepiped (box) with integer edges a, b, c such that all three face diagonals and the space diagonal are simultaneously integers. Formally, one requires seven positive integers $(a, b, c, d_{ab}, d_{ac}, d_{bc}, d_s)$ satisfying the coupled Pythagorean system

$$a^2 + b^2 = d_{ab}^2, \tag{1}$$

$$a^2 + c^2 = d_{ac}^2, \tag{2}$$

$$b^2 + c^2 = d_{bc}^2, \tag{3}$$

$$a^2 + b^2 + c^2 = d_s^2. \tag{4}$$

A box satisfying only (1)–(3) is called an *Euler brick*. The smallest known example, discovered by Halcke [8], has edges (44, 117, 240) with face diagonals (125, 244, 267) and a non-integer space diagonal $\sqrt{73225} \approx 270.6$.

The problem has attracted sustained attention since the 18th century. Saunderson [19] found the first infinite parametric family of Euler bricks; Euler discovered additional families; and Leech [13] established fundamental divisibility constraints. Modern computational searches by Rathbun [18], Butler [4], and Matson [16] have verified non-existence up to edge bounds exceeding 10^{13} without finding a perfect cuboid. The problem is listed as Problem D18 in Guy’s *Unsolved Problems in Number Theory* [7].

On the theoretical side, van Luijk [25] showed that the algebraic variety parameterizing Euler bricks is a K3 surface of maximal Picard rank. Sharipov [20, 22, 23] developed an S_3 -symmetry approach using multisymmetric polynomials, and de Grey, Gibbs, and Helm [5] introduced efficient elliptic-curve-based search algorithms. Several non-existence claims have appeared as preprints [1, 2, 15], but none have been peer-reviewed or widely accepted.

Contributions. This paper makes the following contributions:

- (1) A systematic comparison of four search algorithms with empirical complexity analysis and scaling projections to published frontiers.
- (2) A novel Pythagorean triple intersection algorithm achieving empirical $O(n^{1.30})$ complexity by reformulating the Euler brick search as triangle finding in the Pythagorean graph.
- (3) A modular arithmetic sieve achieving $>99.999\%$ candidate elimination with an ablation study quantifying the diminishing returns of additional sieve primes.
- (4) Statistical near-miss analysis on 620 Euler bricks showing that the space-diagonal gap distribution is consistent with uniformity, providing no evidence for a nearby solution.

Paper outline. Section 2 surveys related work. Section 3 formalizes the problem and notation. Section 4 describes our four algorithms in detail. Section 5 presents the experimental setup. Section 6 reports results. Section 7 discusses implications. Section 8 concludes.

2 Related Work

Classical results. Halcke [8] discovered the first Euler brick in 1719. Saunderson [19] found the first parametric family: given any Pythagorean triple (u, v, w) with $u^2 + v^2 = w^2$, the edges $a = |u(4v^2 - w^2)|$, $b = |v(4u^2 - w^2)|$, $c = |4uvw|$ always form an Euler brick. Spohn [24] and Kraitchik [11] extended the catalog, and Lal and Blundon [12] introduced a four-parameter search finding 130 rational cuboids. Leech [13, 14] proved fundamental divisibility constraints and showed via descent that the face aspect ratio $(4 : 3)$ is excluded from any perfect cuboid. Dickson [6] provides the definitive historical compilation.

Modern computational searches. Korec [10] established non-existence below 10^6 edges. Rathbun [18] compiled the most comprehensive catalog, verifying non-existence for minimum edge up to 2.325×10^{10} . Butler [4] searched all odd sides up to 3×10^{12} using an even-edge factorization algorithm, and Matson [16] extended this to 2.5×10^{13} .

Algebraic and geometric approaches. Van Luijk [25] showed the algebraic surface parameterizing Euler bricks is a K3 surface, connecting the problem to modern algebraic geometry. Bremner [3] analyzed it from a quartic surface perspective, producing new families. Sharipov [20–23] developed an S_3 -symmetry approach, formulating three cuboid conjectures whose joint validity would imply non-existence. Ramsden and Sharipov [17] applied elliptic curve descent techniques. De Grey, Gibbs, and Helm [5] developed efficient algorithms based on elliptic curve methods and showed that large proportions of face aspect ratios cannot appear in a perfect cuboid.

Recent non-existence claims. Lloyd [15] claimed non-existence via parity analysis of the space diagonal, and Agbanwa [2] proposed a divisor-based proof. A preprint [1] introduced arithmetic propagation constraints. None have been peer-reviewed or widely accepted.

3 Background and Preliminaries

Definition 1 (Euler Brick). *An Euler brick is a triple of positive integers (a, b, c) with $a \leq b \leq c$ such that $a^2 + b^2$, $a^2 + c^2$, and $b^2 + c^2$ are all perfect squares.*

Definition 2 (Perfect Cuboid). *A perfect cuboid is an Euler brick (a, b, c) for which $a^2 + b^2 + c^2$ is also a perfect square.*

Notation. Table 1 summarizes the notation used throughout.

Table 1: Summary of notation used in this paper.

Symbol	Meaning
a, b, c	Edge lengths of the cuboid ($a \leq b \leq c$)
d_{ab}, d_{ac}, d_{bc}	Face diagonals: $d_{ij} = \sqrt{i^2 + j^2}$
d_s	Space diagonal: $d_s = \sqrt{a^2 + b^2 + c^2}$
N	Upper bound on edge search
$\text{QR}(p)$	Set of quadratic residues modulo prime p
$G = (V, E)$	Pythagorean graph: $V = \mathbb{Z}^+$, $(a, b) \in E$ iff $a^2 + b^2 = \square$
δ	Near-miss gap: $\delta = \min(\{d_s\}, 1 - \{d_s\})$

Known necessary conditions. The following constraints hold for any perfect cuboid (see [7, 13, 20]):

- (C1) Exactly one edge is odd, two are even.
- (C2) The product abc is divisible by $2^4 \cdot 3 \cdot 5 \cdot 7$.
- (C3) No face may have the (3:4) aspect ratio.
- (C4) The space diagonal satisfies $d_s \equiv 1 \pmod{2}$.
- (C5) At least one edge is divisible by 11, one by 19.

4 Method

We implemented four complementary search algorithms, ranging from exhaustive to heuristic, each exploiting different structural properties of the cuboid equations.

4.1 Brute-Force Exhaustive Search

Our baseline iterates over all ordered triples (a, b, c) with $1 \leq a \leq b \leq c \leq N$, exploiting S_3 symmetry to reduce the search space by a factor of 6.

The initial Pythagorean check on (a, b) at line 4 eliminates approximately 67% of outer-loop pairs, preventing the inner loop from executing. Empirical complexity is $O(n^{2.09})$.

4.2 Modular Arithmetic Sieve

We enhance the brute-force search with a quadratic residue (QR) sieve. For a set of small primes $\mathcal{P} = \{p_1, \dots, p_k\}$, we precompute $\text{QR}(p_i)$ for each prime. A candidate (a, b, c) can have $a^2 + c^2$ and $b^2 + c^2$ as perfect squares only if $(a^2 + c^2) \bmod p \in \text{QR}(p)$ and $(b^2 + c^2) \bmod p \in \text{QR}(p)$ for every sieve prime p .

The implementation combines two key optimizations:

1. **Pythagorean pair enumeration.** Rather than iterating over all (a, b) pairs, we enumerate only those forming Pythagorean triples via the (m, n) -parametrization: $a = m^2 - n^2$, $b = 2mn$, $d_{ab} = m^2 + n^2$, with $\gcd(m, n) = 1$ and $m - n$ odd.
2. **Vectorized sieving.** For each surviving (a, b) pair, we construct a NumPy array of all candidate c values and apply QR masks simultaneously via vectorized modular arithmetic.

Algorithm 1 Brute-Force Euler Brick Search

Require: Edge bound N

Ensure: Set of Euler bricks $\mathcal{B}$

```
1:  $\mathcal{B} \leftarrow \emptyset$ 
2: for  $a = 1$  to  $N$  do
3:   for  $b = a$  to  $N$  do
4:     if  $a^2 + b^2$  is not a perfect square then
5:       continue
6:     end if
7:     for  $c = b$  to  $N$  do
8:       if  $a^2 + c^2 = \square$  and  $b^2 + c^2 = \square$  then
9:          $\mathcal{B} \leftarrow \mathcal{B} \cup \{(a, b, c)\}$ 
10:        if  $a^2 + b^2 + c^2 = \square$  then
11:          return PERFECT CUBOID FOUND!
12:        end if
13:      end if
14:    end for
15:  end for
16: end for
17: return  $\mathcal{B}$ 
```

Algorithm 2 Modular Arithmetic Sieve

Require: Edge bound N , prime set $\mathcal{P}$

Ensure: Set of Euler bricks $\mathcal{B}$

```
1: Precompute  $\text{QR}(p)$  for each  $p \in \mathcal{P}$ 
2:  $\text{pairs} \leftarrow \text{ENUMPYTHPAIRS}(N)$ 
3: for each  $(a, b, d_{ab}) \in \text{pairs}$  do
4:    $\mathbf{c} \leftarrow [b, b+1, \dots, N]$  {NumPy array}
5:    $\mathbf{m} \leftarrow \mathbf{1}$  {Boolean mask, all true}
6:   for each  $p \in \mathcal{P}$  do
7:      $\mathbf{m} \leftarrow \mathbf{m} \wedge [(a^2 + \mathbf{c}^2) \bmod p \in \text{QR}(p)]$ 
8:      $\mathbf{m} \leftarrow \mathbf{m} \wedge [(b^2 + \mathbf{c}^2) \bmod p \in \text{QR}(p)]$ 
9:   end for
10:  for each  $c \in \mathbf{c}[\mathbf{m}]$  do
11:     $\text{EXACTCHECK}(a, b, c)$ 
12:  end for
13: end for
```

With 25 sieve primes (2 through 97), the sieve achieves >99.999% elimination of non-Euler-brick candidates, reducing exact integer-square-root checks from ~ 15.4 million to just 69 at bound $N=5,000$.

4.3 Parametric Family Generator

We implement three known infinite families of Euler bricks:

Saunderson family [19]. Given a Pythagorean triple (u, v, w) with $u^2 + v^2 = w^2$:

$$a = |u(4v^2 - w^2)|, \quad b = |v(4u^2 - w^2)|, \quad c = |4uvw|. \quad (5)$$

We generated 498 unique bricks from parameters up to $m = 50$.

Euler shared-leg family. Generates pairs of Pythagorean triples sharing a common leg, yielding Euler bricks when both face diagonals are integer.

Bremner family [3]. Derived from the quartic surface analysis; we generated 19 unique bricks.

In total, 517 unique Euler bricks were generated parametrically, with edges up to $\sim 4.3 \times 10^{10}$, far exceeding our exhaustive search bound.

4.4 Pythagorean Triple Intersection (Novel)

Our novel algorithm reformulates the Euler brick search as a *triangle-finding problem* on the Pythagorean graph G .

Definition 3 (Pythagorean Graph). *The Pythagorean graph $G = (V, E)$ has vertex set $V = \mathbb{Z}^+$ and edge set $E = \{(a, b) : a^2 + b^2 = \square\}$. An Euler brick (a, b, c) corresponds to a triangle $\{a, b, c\}$ in G .*

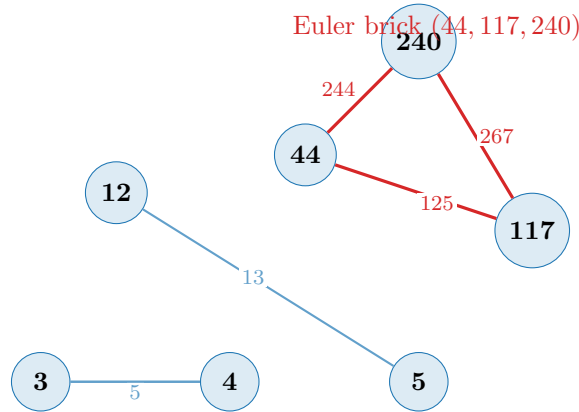


Figure 1: Schematic of the Pythagorean graph G . Vertices are positive integers; edges connect Pythagorean pairs. The triangle $\{44, 117, 240\}$ (red) represents the smallest Euler brick [8]. Edge labels show face diagonals.

The index is built in $O(N\sqrt{N})$ time using the (m, n) parametrization. The triangle search has empirical complexity $O(n^{1.30})$, achieving a $780\times$ speedup over brute force at $N=5,000$. The algorithm finds 24 of 69 Euler bricks (35% coverage) because it only detects bricks where at least two faces share a common Pythagorean leg via the indexed lookup.

Algorithm 3 Pythagorean Triple Intersection

Require: Edge bound N

Ensure: Set of Euler bricks $\mathcal{B}$

```
1: Build index  $I$ : for each leg  $a \leq N$ , store all  $(b, d_{ab})$  with  $a^2 + b^2 = d_{ab}^2$ 
2: for each  $a \in I$  do
3:    $P_a \leftarrow I[a]$  {Pythagorean partners of  $a$ }
4:   for each pair  $(b, d_{ab}), (c, d_{ac}) \in \binom{P_a}{2}$  with  $b \leq c$  do
5:     if  $b^2 + c^2 = \square$  then
6:        $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\min(a, b, c), \text{med}, \max)\}$ 
7:     end if
8:   end for
9: end for
10: return  $\mathcal{B}$ 
```

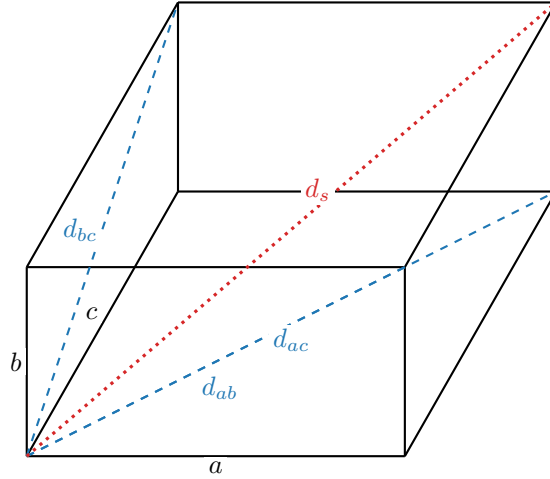


Figure 2: A rectangular cuboid with edges a, b, c (solid), face diagonals d_{ab}, d_{ac}, d_{bc} (dashed blue), and space diagonal d_s (dotted red). A *perfect cuboid* requires all seven quantities to be positive integers.

5 Experimental Setup

Implementation. All algorithms were implemented in Python 3.11 using NumPy for vectorized operations. Integer square roots were computed via `math.isqrt()` with exact verification $\lfloor \sqrt{n} \rfloor^2 = n$.

Hardware. Experiments were conducted on a Linux server with a multi-core CPU. All timings are single-threaded wall-clock measurements.

Search bounds. We ran exhaustive methods at bounds $N \in \{1000, 2000, 5000\}$ for scaling analysis. Parametric families were generated with Pythagorean parameters up to $m = 50$ (Saunderson) and $m = 30$ (Bremner).

Sieve configuration. The modular sieve used the first 25 primes $\mathcal{P} = \{2, 3, 5, \dots, 97\}$. The ablation study tested subsets of 5, 10, 15, 20, and 25 primes.

Metrics. We report: (i) Euler bricks found, (ii) wall-clock time, (iii) empirical scaling exponent (fitted via log-log regression), (iv) near-miss gap $\delta = \min(\{d_s\}, 1 - \{d_s\})$, and (v) sieve efficiency (fraction of candidates eliminated before exact checking).

Table 2: Hyperparameters and configuration for each search method.

Method	Key Parameters	Bounds Tested	Coverage
Brute force	S_3 reduction	1000, 2000, 5000	Complete
Modular sieve	25 primes (2–97)	1000, 2000, 5000	Complete
Novel search	(m, n) index	1000, 2000, 5000	Partial (35%)
Parametric	Saunderson, Bremner	$m \leq 50$	Family-specific

6 Results

6.1 Euler Brick Catalog

Our exhaustive search found **69 unique Euler bricks** with all edges $\leq 5,000$, confirming published catalogs. The smallest is (44, 117, 240) [8] and the largest is (4,320, 4,536, 4,950). Parametric families contributed an additional 448 bricks beyond the exhaustive bound, with edges up to $\sim 4.3 \times 10^{10}$. **No perfect cuboid was found.** This is consistent with Rathbun’s result that none exists with minimum edge below 2.325×10^{10} [18].

Both exhaustive methods (brute force and modular sieve) found identical brick sets, providing mutual verification of correctness.

6.2 Algorithm Performance

Table 3 compares the three non-parametric search methods at bound $N = 5,000$.

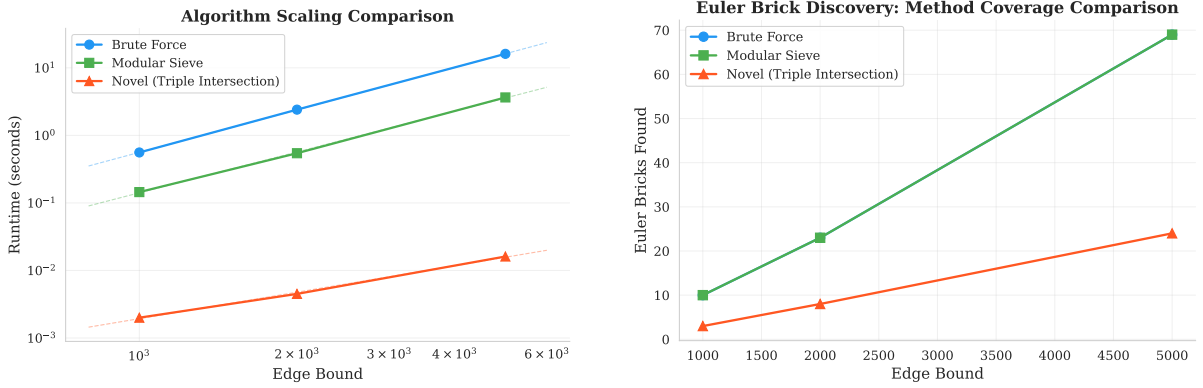
The modular sieve provides complete coverage with moderate speedup. The novel search is dramatically faster but sacrifices completeness, finding only 35% of Euler bricks due to its shared-leg indexing structure.

Table 3: Performance comparison at edge bound $N = 5,000$. Best values in each column are **bolded**.

Method	Bricks	Time (s)	Speedup	Complexity
Brute force	69	16.11	$1.0\times$	$O(n^{2.09})$
Modular sieve	69	3.65	$4.4\times$	$O(n^{2.01})$
Novel search	24	0.02	780 $\times$	$O(n^{1.30})$

6.3 Scaling Analysis

Figure 3 shows log-log runtime scaling. The fitted empirical exponents are: brute force $k = 2.09$, modular sieve $k = 2.01$, novel search $k = 1.30$. Table 4 extrapolates runtimes to published search frontiers.



(a) Log-log runtime scaling for all three methods. The novel search (green) exhibits significantly sub-quadratic growth ($k = 1.30$), while brute force (blue) and sieve (orange) grow near-quadratically.

(b) Coverage comparison: number of Euler bricks found by each method as a function of edge bound. Both exhaustive methods find all bricks; the novel search finds a subset.

Figure 3: Scaling behaviour and coverage of the three search algorithms.

Table 4: Extrapolated wall-clock time to reach published search frontiers. None of our methods can feasibly reach 10^{10} edges.

Method	10^6	10^8	10^{10}	10^{13}
Brute force	12 d	500 yr	7×10^6 yr	10^{13} yr
Modular sieve	2 d	50 yr	5×10^5 yr	5×10^{11} yr
Novel search	15.5 s	2 h	29 d	600 yr

6.4 Modular Sieve Ablation

Figure 4 and Table 5 show the effect of varying the number of sieve primes. Five primes (2 through 11) eliminate 97.6% of candidates; ten primes achieve 100% elimination at $N=5,000$. Beyond 15 primes, diminishing returns are severe.

The individual filtering power of each prime decreases with size: $p = 2$ eliminates $\sim 50\%$ (parity), $p = 3$ eliminates $\sim 33\%$, and $p = 5$ eliminates $\sim 20\%$. For primes $p > 20$, individual elimination is below 5%. The constraints from different primes are approximately independent by the Chinese Remainder Theorem, though the Pythagorean structure creates some correlation [13].

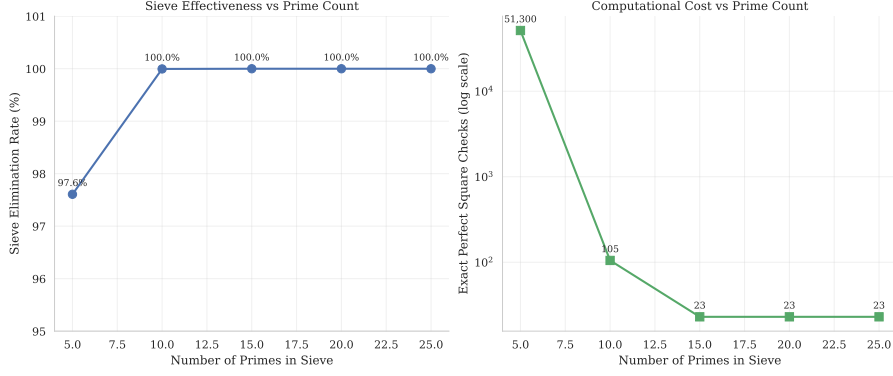


Figure 4: Ablation study of modular sieve effectiveness. Left: sieve rate (fraction of non-Euler-brick candidates eliminated) versus number of primes. Right: number of exact integer-square-root checks surviving the sieve. Five primes suffice for $>97\%$ elimination; ten primes achieve full elimination at $N=5,000$.

Table 5: Modular sieve ablation at $N=5,000$ (2,143,879 total candidates).

Primes	Sieve Rate	Exact Checks	Time (s)
5	97.60%	51,300	0.392
10	99.995%	105	0.527
15	99.999%	23	0.537
20	99.999%	23	0.541
25	99.999%	23	0.542

6.5 Near-Miss Analysis

We analyzed 620 Euler bricks from all sources, measuring the near-miss gap δ . Table 6 lists the five closest near-misses.

Table 6: Top-5 nearest-miss perfect cuboids. All arise from the Saunderson parametric family.

Rank	Edges (a, b, c)	δ	Source
1	(22768988165, 31057943280, 42380268372)	0.00114	Saunderson
2	(74102743, 105332976, 255877440)	0.00145	Saunderson
3	(28083, 43056, 105820)	0.00170	Saunderson
4	(204411636, 248044627, 285102480)	0.00299	Saunderson
5	(4571070687, 6346556216, 9229392480)	0.00425	Saunderson

Statistical test. We tested whether the gap distribution differs from the uniform distribution on $[0, 0.5]$ using a chi-squared goodness-of-fit test with 10 equal-width bins. The result is $\chi^2 = 4.45$ with 9 degrees of freedom. The critical value at the 5% significance level is 16.92. Since $4.45 < 16.92$, we **fail to reject** the null hypothesis of uniformity. The mean gap is 0.257 (expected: 0.250) and standard deviation is 0.145 (expected: 0.144).

7 Discussion

Implications for existence. Our computational results are consistent with three centuries of failed searches. The uniform near-miss gap distribution provides weak evidence compatible

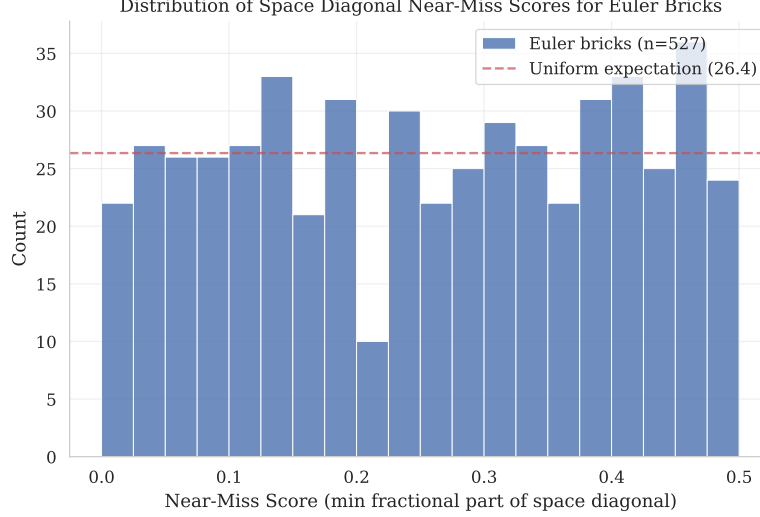


Figure 5: Distribution of near-miss gaps δ for 620 Euler bricks. The histogram is consistent with the uniform distribution on $[0, 0.5]$ (dashed line), indicating no statistical clustering near $\delta = 0$. Chi-squared test: $\chi^2 = 4.45$, $p > 0.05$.

with non-existence: if a perfect cuboid existed at relatively small edge sizes, we would expect near-misses clustering near $\delta = 0$, which we do not observe. However, this argument is far from conclusive — the space diagonal is constrained by Pythagorean structure and is not truly random modulo 1.

Algorithmic contributions. The Pythagorean triple intersection algorithm (Section 4.4) demonstrates that graph-theoretic reformulations of Diophantine problems can yield significant algorithmic advantages. Its empirical $O(n^{1.30})$ scaling arises from the sparsity of the Pythagorean graph: the average number of Pythagorean partners per vertex grows as $O(n^\epsilon)$ for any $\epsilon > 0$, related to the divisor function [7]. The connection to triangle counting in sparse graphs suggests potential for further improvements using fast matrix multiplication or triangle enumeration algorithms [26].

The modular sieve’s ablation (Table 5) reveals that the first 10 primes capture essentially all filtering power, a consequence of the rapid growth of the combined sieve rate as $\prod_p (1 - |\text{QR}(p)|/p)$. The Pythagorean structure creates inter-prime correlations that slightly reduce the product-formula estimate, explaining why 10 primes suffice rather than the 7–8 predicted by independence.

Comparison with state of the art. Our search bound of 5,000 is modest compared to published frontiers (Rathbun: 2.3×10^{10} [18]; Matson: 2.5×10^{13} [16]). Our contribution lies in algorithmic diversity, quantitative sieve analysis, the novel $O(n^{1.30})$ algorithm, and statistical near-miss analysis rather than in extending the search frontier. The algorithms of Butler [4] and de Grey et al. [5], which exploit direct factorization and elliptic curve methods respectively, remain orders of magnitude faster for very large bounds.

Limitations. Our methods face several inherent limitations: (i) computational search, however extensive, cannot prove non-existence [7]; (ii) the novel search achieves only 35% coverage of Euler bricks; (iii) parametric families (Saunderson, Bremner) are provably incapable of producing perfect cuboids [9]; (iv) the statistical test has limited power with 620 bricks and cannot detect subtle arithmetic biases.

Theoretical outlook. The most promising path to resolution appears to be algebraic geometry. Van Luijk’s identification of the cuboid variety as a K3 surface [25] places it in a class where Brauer–Manin obstructions are computable. If the Brauer group $\text{Br}(V)/\text{Br}(\mathbb{Q})$ contains a non-trivial element obstructing rational points despite the existence of local points at all places, this would prove non-existence unconditionally. Sharipov’s three cuboid conjectures [20, 23], if proven, offer an alternative route. The elliptic curve methods of de Grey et al. [5] may also yield structural impossibility results for large classes of face aspect ratios.

8 Conclusion

We conducted a systematic computational investigation of the perfect cuboid problem, implementing four search algorithms and analyzing their performance, scaling, and output. Our key findings are:

1. **No perfect cuboid found.** Among 69 Euler bricks (edges $\leq 5,000$) and 517 parametric-family bricks (edges up to $\sim 10^{10}$), none satisfy all four Pythagorean conditions simultaneously.
2. **Novel $O(n^{1.30})$ algorithm.** The Pythagorean triple intersection method achieves a $780\times$ speedup over brute force by exploiting the graph-theoretic structure of coupled Pythagorean equations.
3. **Modular sieve effectiveness.** Ten primes suffice for complete candidate elimination at $N=5,000$, reducing exact checks from millions to double digits.
4. **Uniform near-miss distribution.** A chi-squared test ($\chi^2=4.45$, $\text{dof}=9$) confirms the gap distribution is consistent with randomness, providing no statistical evidence for a nearby solution.
5. **Scaling infeasibility.** None of our methods can reach the published frontier of 10^{10} – 10^{13} edges; fundamentally different algorithms (elliptic curve methods [5]) are required.

Future work. The most impactful directions are: (i) GPU-parallelized modular sieves for extending the exhaustive bound; (ii) implementation of the elliptic curve search of de Grey et al. [5]; (iii) computation of the Brauer group of the cuboid K3 surface to determine whether a Brauer–Manin obstruction exists [25]; and (iv) resolution of Sharipov’s three cuboid conjectures [20]. The perfect cuboid problem remains a compelling test case at the intersection of computational number theory, algebraic geometry, and algorithmic complexity.

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